Shocks, Weak Solutions, and Entropy: An Exposition of Burgers' Equation

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1 Motivation

1.1 The question, and what is original here

Question. *Why does the inviscid Burgers' equation need an entropy condition, and why is that the right condition?* The answer developed here is a chain: smooth data develop singularities in finite time (Sections 2–3); the weakest sensible notion of solution then admits discontinuities (Section 4) but is non-unique (Section 5); an entropy condition restores uniqueness (Sections 6–7); and the vanishing-viscosity limit shows this condition is the physically correct one (Section 8).

What is standard, and what is not. The theory in Sections 2–7 and the statements in Sections 8.1–8.3 are standard (see [5, §3.4]); the aim is a careful, complete account with every verification done explicitly. The parts that go beyond a direct exposition are *computations and observations*, labelled as such rather than as new theorems: (i) the worked example $u_0 = -\arctan x$ is carried through breakdown to an explicit entropy solution with an explicit shock strength; the Lax condition is derived from viscosity (Proposition 8.6); the layer part of the example's vanishing-viscosity error is *predicted* (a viscous shock of any strength has L^1 distance $4\ln 2\,\nu$ from the sharp shock) and checked numerically, including its independence of the shock strength (Section 8.4); (ii) the Godunov scheme is implemented and its convergence measured, including a grid-alignment effect on the shock error, and a modified-equation test of the heuristic that its numerical diffusion acts as a vanishing viscosity (Section 8.5); (iii) the N-wave asymptotics are checked numerically, including a counterexample to the tempting but incorrect identification of the N-wave constants with the positive and negative masses of the data (Section 8.6).

1.2 Burgers' equation as a model problem

Burgers' equation was proposed by J. M. Burgers as a one-dimensional caricature of turbulence [2]; the equation had appeared earlier in work of Bateman [1]. It retains the two competing effects that make the Navier–Stokes equations hard to analyse: nonlinear advection, which steepens a velocity profile, and viscous diffusion, which smooths it. Writing $u(x,t)$ for the velocity and $\nu > 0$ for the viscosity, the *viscous Burgers' equation* is

$$u_t + u\,u_x = \nu\,u_{xx}. \tag{1}$$

This should be read as a model rather than a derivation: it has the form of the Navier–Stokes momentum equation with the pressure term removed. (In higher dimensions the pressure is determined implicitly by the incompressibility constraint $\nabla\cdot u = 0$, and this nonlocal coupling is a main source of difficulty. In one dimension incompressibility would force $u_x = 0$, so there is no one-dimensional analogue of it and the pressure is simply omitted.) Setting $\nu = 0$ gives the *inviscid* equation

$$u_t + u\,u_x = 0, \tag{2}$$

which is the main object of study in Sections 2–7, before viscosity is reintroduced in Section 8. What remains is a single scalar equation depending only on local values of u, but one that retains the nonlinear term $u\,u_x$ responsible for the spontaneous formation of discontinuities studied from Section 3 onward.

1.3 Burgers' equation as a scalar conservation law

Equations (2) and (1) are instances of a *scalar conservation law*,

$$u_t + f(u)_x = 0, \tag{3}$$

with flux f, together with its viscous regularisation $u_t + f(u)_x = \nu\, u_{xx}$. For Burgers' equation,

$$f(u) = \frac{u^2}{2}, \tag{4}$$

so $f(u)_x = u\, u_x$ recovers (2). Such equations describe a locally conserved quantity transported at state-dependent speed $f'(u)$, and arise throughout gas dynamics and traffic flow [11]. Burgers' equation is the simplest nontrivial case (a linear f gives only the trivial transport equation $u_t + c\, u_x = 0$), and is the standard first example for the general theory of characteristics, shocks, weak solutions, and entropy conditions developed in Sections 2–6.

1.4 Why study Burgers' equation

Three features make Burgers' equation worth studying in its own right. First, it is the simplest equation whose smooth solutions can lose smoothness in finite time despite smooth data and no forcing; because it is scalar and one-dimensional, this can be analysed completely. (This is a model of nonlinear steepening and its regularisation by viscosity, not of the mechanisms behind the regularity questions for three-dimensional Euler and Navier–Stokes, which are still open: the scalar one-dimensional structure of Burgers' equation — a maximum principle, no pressure, no vortex stretching — makes the underlying mathematics quite different.) Second, once weak solutions are admitted, it exhibits genuine non-uniqueness (Section 5), requiring an entropy condition (Section 6) to restore well-posedness — a phenomenon common to conservation laws in general, studied here with minimal technical overhead. Third, the viscous equation (1) is one of the few nonlinear PDEs solvable in closed form, via the Cole–Hopf transformation (Section 8), allowing an explicit comparison between the viscous and inviscid theories.

2 Method of Characteristics

This section derives an implicit solution formula for the inviscid Burgers' equation via the method of characteristics. The formula's eventual breakdown is what forces the more general framework of Sections 3–6.

2.1 Setup

Consider the initial value problem

$$\begin{cases} u_t + u\, u_x = 0, & x \in \mathbb{R},\ t > 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \tag{5}$$

with $u_0 \in C^1(\mathbb{R})$ in Sections 2 and 3. (Later sections use discontinuous data such as the Riemann data (26); there (5) is understood in the weak sense of Definition 4.1, and the classical theory of this section does not apply directly.) The method of characteristics looks for curves $x(t)$ along which the PDE reduces to an ODE for $u(x(t), t)$.

2.2 Derivation of the characteristic equations

For a solution u of (5) and a curve $x(t)$, the chain rule gives

$$\frac{d}{dt}\, u(x(t), t) = u_x(x(t), t)\, \dot{x}(t) + u_t(x(t), t). \tag{6}$$

Since $u_t = -u\, u_x$ on solutions, choosing $x(t)$ to satisfy

$$\dot{x}(t) = u(x(t), t) \tag{7}$$

reduces (6) to

$$\frac{d}{dt}\, u(x(t), t) = 0. \tag{8}$$

Curves satisfying (7) are *characteristics*, and u is constant along each one.

2.3 Characteristics are straight lines

Proposition 2.1 (Characteristics are straight lines)**.** *Let u be a C^1 solution of (5) on $\mathbb{R} \times [0, T)$, and let $x(t)$ solve (7) with $x(0) = x_0$. Then for $0 \le t < T$,*

$$u(x(t), t) = u_0(x_0), \qquad x(t) = x_0 + u_0(x_0)\, t. \tag{9}$$

Proof. By (8), $u(x(t),t) \equiv u(x_0,0) = u_0(x_0)$. Substituting this constant into (7) gives $\dot{x}(t) = u_0(x_0)$, which integrates to the stated formula for $x(t)$. □

Thus any classical solution is completely determined by (9), reducing its construction to inverting $x_0 \mapsto x = x_0 + u_0(x_0)\,t$ at each fixed t.

2.4 The implicit solution formula

Theorem 2.2 (Implicit solution via characteristics)**.** *Let $u_0 \in C^1(\mathbb{R})$. For each $t \geq 0$ such that $\Phi_t : \mathbb{R} \to \mathbb{R}$, $\Phi_t(x_0) = x_0 + u_0(x_0)\,t$, is a bijection with C^1 inverse, the function u defined implicitly by*

$$u(x,t) = u_0(x_0), \qquad \text{where } x = x_0 + u_0(x_0)\,t, \tag{10}$$

is the unique C^1 solution of (5) at time t.

Proof. Well-definedness. Bijectivity of Φ_t means $u(x,t) = u_0(\Phi_t^{-1}(x))$ is well defined, and C^1 since Φ_t^{-1} and u_0 are.

The PDE holds. Write $x_0 = x_0(x,t) = \Phi_t^{-1}(x)$. Differentiating $x = x_0 + u_0(x_0)\,t$ in x and in t gives

$$\frac{\partial x_0}{\partial x} = \frac{1}{1 + u_0'(x_0)\,t}, \qquad \frac{\partial x_0}{\partial t} = \frac{-u_0(x_0)}{1 + u_0'(x_0)\,t}, \tag{11}$$

both well defined since $\Phi_t'(x_0) = 1 + u_0'(x_0)t \neq 0$. By the chain rule applied to $u(x,t) = u_0(x_0(x,t))$,

$$u_x = \frac{u_0'(x_0)}{1 + u_0'(x_0)\,t}, \qquad u_t = \frac{-u_0(x_0)\,u_0'(x_0)}{1 + u_0'(x_0)\,t}, \tag{12}$$

so $u_t + u\,u_x = 0$, and $u(x,0) = u_0(x)$ since $\Phi_0 = \mathrm{id}$.

Uniqueness. Let v be any C^1 solution of (5) on $\mathbb{R} \times [0,t]$, and fix $x \in \mathbb{R}$. Solve backwards in time the ODE $\dot{\xi}(s) = v(\xi(s),s)$, $\xi(t) = x$ (the right-hand side is C^1, hence locally Lipschitz, so there is a unique maximal solution). By (8), v is constant along ξ, hence so is $\dot{\xi}$: ξ is a straight line with bounded slope $v(x,t)$. It therefore stays bounded and cannot blow up, so it extends back to $s = 0$. Setting $x_0 = \xi(0)$ we get $v(x,t) = v(x_0,0) = u_0(x_0)$ and $x = x_0 + u_0(x_0)t$. Thus every point (x,t) lies on a characteristic coming from $t = 0$, with $x_0 = \Phi_t^{-1}(x)$, and $v(x,t) = u_0(\Phi_t^{-1}(x)) = u(x,t)$. □

Corollary 2.3 (Local existence)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \inf_{x_0} u_0'(x_0) > -\infty$, and set $T = \infty$ if $m \geq 0$ and $T = -1/m$ if $m < 0$. Then (5) has a unique C^1 solution on $\mathbb{R} \times [0,T)$, given by (10).*

Proof. Fix $t \in [0,T)$ and put $c_t := 1 + \min(m,0)\,t$, so $c_t > 0$ and $\Phi_t'(x_0) = 1 + u_0'(x_0)t \geq c_t$ for every x_0. Three consequences follow. (i) Φ_t is strictly increasing, hence injective. (ii) By the mean value theorem $\Phi_t(x_0) - \Phi_t(0) \geq c_t x_0$ for $x_0 \geq 0$ and $\leq c_t x_0$ for $x_0 \leq 0$, so $\Phi_t(x_0) \to \pm\infty$ as $x_0 \to \pm\infty$; by continuity Φ_t is *onto* $\mathbb{R}$. (iii) Φ_t' never vanishes, so the inverse function theorem gives a C^1 inverse, depending C^1 on t by the implicit function theorem applied to $x_0 + u_0(x_0)t - x = 0$. Theorem 2.2 therefore applies at every $t < T$. □

Remark 2.4 (Why the hypothesis on u_0' matters). Strict monotonicity of Φ_t alone would not give surjectivity; it is the *uniform* lower bound $\Phi_t' \geq c_t > 0$ that does. The hypothesis cannot be dropped: for $u_0(x) = x^2$ we have $u_0'(x) = 2x$, unbounded below, and $\Phi_t(x) = x + tx^2$ fails to be injective for every $t > 0$ (since $\Phi_t'(x) < 0$ for $x < -1/(2t)$), i.e. the classical solution breaks down immediately, in agreement with Proposition 3.1 read with $m = -\infty$.

The bound $T \leq -1/m$ is not an artefact of the proof: as $t \uparrow -1/m$, $\sup_x |u_x(\cdot,t)| \to \infty$, by (14) along characteristics with $u_0' \to m$. When the infimum m is attained, Proposition 3.1 shows this is exactly where the classical solution first fails to exist; when it is not attained the derivative blows up as $t \uparrow -1/m$ without a first point of degeneracy.

2.5 Worked example

Let $u_0(x) = -\arctan(x)$, so $u_0'(x) = -1/(1+x^2) \in [-1,0)$. By Theorem 2.2, the solution satisfies

$$u(x,t) = -\arctan(x_0), \qquad x = x_0 - \arctan(x_0)\,t, \tag{13}$$

valid while $\Phi_t'(x_0) = 1 - t/(1+x_0^2)$ is positive for all x_0. Since $1/(1+x_0^2) \leq 1$ with equality only at $x_0 = 0$, $\Phi_t' > 0$ everywhere exactly when $t < 1$. At $t = 1$ the derivative vanishes for the first time, at $x_0 = 0$: the characteristic through the origin, carrying the steepest slope $u_0'(0) = -1$, is the first to degenerate. At $t = 1$ the map $\Phi_1(x_0) = x_0 - \arctan x_0$ is still strictly increasing but its inverse fails to be C^1 at 0; for $t > 1$, $\Phi_t' < 0$ on an interval around $x_0 = 0$, Φ_t is no longer injective, and characteristics have crossed.

Remark 2.5 (What happens beyond T). Once two characteristics carrying different values of u_0 meet, formula (10) assigns u two values at once, so no single-valued classical solution can exist beyond this time. Making sense of the equation past this point is the subject of Sections 3 and 4.

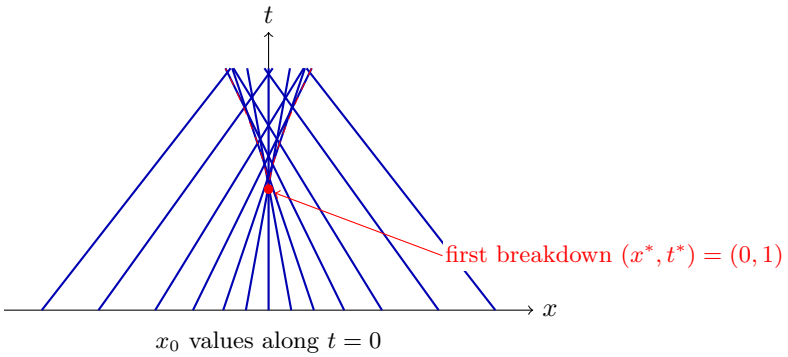


Figure 1: Characteristics $x = x_0 - \arctan(x_0)\, t$ for $u_0 = -\arctan x$, together with their envelope (red, dashed), which has a cusp at $(x^*, t^*) = (0, 1)$. Where u_0 is decreasing, characteristics converge; the first breakdown time is $T^* = -1/\min u_0' = 1$ (Corollary 2.3), computed precisely in Section 3. Inside the cusp ($t > 1$) three characteristics pass through each point.

3 Breakdown of Classical Solutions

This section computes the exact time and location at which the classical solution of Section 2 first breaks down. What breaks down is the *smoothness* of u (its slope becomes infinite); the discontinuities (shocks) that replace it afterwards belong to the weak solutions of Sections 4–7.

3.1 Blow-up of the derivative

From the proof of Theorem 2.2,

$$u_x(x,t) = \frac{u_0'(x_0)}{1 + u_0'(x_0)\, t}, \qquad x = x_0 + u_0(x_0)\, t. \tag{14}$$

If $u_0'(x_0) \geq 0$, the denominator stays ≥ 1 and u_x remains bounded along that characteristic for all $t \geq 0$. If $u_0'(x_0) < 0$, the denominator vanishes at the finite time $t = -1/u_0'(x_0)$, where (14) shows $|u_x| \to \infty$: the solution value stays bounded, but its slope becomes vertical.

3.2 The envelope of the characteristics

The characteristics form a one-parameter family of lines

$$x(x_0, t) = x_0 + u_0(x_0)\, t, \qquad x_0 \in \mathbb{R}. \tag{15}$$

Neighbouring members of this family meet where $\partial x/\partial x_0 = 0$, i.e. on the envelope of the family, which is found by solving (15) together with

$$1 + u_0'(x_0)\, t = 0. \tag{16}$$

This is exactly the vanishing of $\Phi_t'(x_0)$, so the envelope condition coincides with the derivative blow-up condition above.

3.3 The breakdown time

Proposition 3.1 (Breakdown time)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \min_{x_0} u_0'(x_0) < 0$, attained at $x_0 = a$ (if the minimum is attained at several points, a is any one of them). Then the classical solution exists and is C^1 on $\mathbb{R} \times [0, T^*)$ for*

$$T^* = \frac{-1}{m}, \tag{17}$$

and fails to extend as a single-valued C^1 function beyond T^*. The first breakdown occurs at

$$x^* = a + u_0(a)\, T^*, \qquad t^* = T^*. \tag{18}$$

Proof. Since $s \mapsto -1/s$ is strictly increasing on $(-\infty, 0)$ and $u_0'(x_0) \geq m$ everywhere, $-1/u_0'(x_0) \geq -1/m = T^*$ for every x_0 with $u_0'(x_0) < 0$, with equality exactly at the minimisers of u_0' (in particular at $x_0 = a$). So T^* is the smallest time at which any characteristic meets a neighbour: $\Phi_t'(x_0) > 0$ for all x_0 throughout $[0, T^*)$, and by Corollary 2.3 (with m the infimum of u_0') this gives a classical solution there, while $\Phi_{T^*}'(a) = 0$ makes Φ_{T^*} fail to be locally invertible at a, so u_x is unbounded near (x^*, t^*) by (14). Substituting $x_0 = a$, $t = T^*$ into (15) gives x^*. □

Remark 3.2 (Singularity versus shock). At (x^*, t^*) the solution u is still bounded and continuous; only its slope is infinite (a *gradient catastrophe*). No discontinuity has yet formed. Proposition 3.1 therefore describes the breakdown of the classical solution, not the construction of a shock; the latter requires the weak formulation of Section 4.

3.4 Worked example, continued

For $u_0(x) = -\arctan(x)$, $u_0'(x_0) = -1/(1 + x_0^2)$ is minimised at $a = 0$ with $m = -1$, so Proposition 3.1 gives $T^* = 1$, matching Section 2.5. The breakdown point is

$$x^* = 0 + (-\arctan(0)) \cdot 1 = 0, \tag{19}$$

so the singularity first appears at $(x^*, t^*) = (0, 1)$, matching Figure 1. Since this u_0 is odd and bounded, the entropy solution (which exists by the Lax–Oleinik construction [5, §3.4.2]) is unique (Theorem 6.7), and $v(x, t) = -u(-x, t)$ is again an entropy solution with the same data, so the solution remains odd in x; hence for $t > 1$ the discontinuity that replaces the classical solution sits at $x = 0$ and, by the Rankine–Hugoniot speed (25), is stationary (speed $(u_\ell + u_r)/2 = 0$). Its strength and the full solution for $t > 1$ are computed explicitly in Section 8.4.

3.5 From breakdown to weak solutions

Proposition 3.1 shows the classical formulation of (5) cannot describe decreasing initial data beyond T^*, even though nothing in the underlying physics suggests the evolution should stop. Section 4 resolves this by weakening the notion of solution: rather than requiring $u_t + u\,u_x = 0$ pointwise, u need only satisfy an integral identity against smooth test functions, which admits discontinuous solutions for all $t \geq 0$.

4 Weak Solutions and the Integral Form

Since classical solutions of (5) cannot be extended past the breakdown time T^* of Proposition 3.1, this section introduces a weaker notion of solution that admits discontinuities, and derives the condition such discontinuities must satisfy. The two verifications needed later — that a piecewise smooth function is a weak solution (Proposition 4.5), and what a jump must satisfy (Theorem 4.4) — are done *once*, here, and reused in Sections 5 and 7.

4.1 The integral formulation

Let u be a C^1 solution of $u_t + f(u)_x = 0$ on $\mathbb{R} \times (0, \infty)$ with $f(u) = u^2/2$ and $u(\cdot, 0) = u_0$, and let $\varphi \in C_c^1(\mathbb{R} \times [0, \infty))$ be any test function, compactly supported but not required to vanish at $t = 0$. Multiplying the PDE by φ, integrating over $\mathbb{R} \times (0, \infty)$, and integrating by parts in both x and t (using that φ has compact support, so all boundary terms at spatial infinity vanish) gives

$$0 = \int_0^\infty \int_{\mathbb{R}} \big(u_t + f(u)_x\big)\varphi \, dx\, dt = -\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt - \int_{\mathbb{R}} u_0(x)\,\varphi(x, 0)\, dx, \tag{20}$$

where the boundary term at $t = 0$ survives because $\varphi(x, 0)$ need not vanish. Equation (20) makes sense whenever $u \in L^\infty$, with no differentiability required of u at all: this motivates the following definition.

Definition 4.1 (Weak solution)**.** Let $u_0 \in L^\infty(\mathbb{R})$. A function $u \in L^\infty(\mathbb{R} \times (0, \infty))$ is a *weak solution* of (5) (with data u_0) if

$$\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt + \int_{\mathbb{R}} u_0(x)\,\varphi(x, 0)\, dx = 0 \tag{21}$$

for every $\varphi \in C_c^1(\mathbb{R} \times [0, \infty))$.

Conventions. Test functions are always taken in $C_c^1(\mathbb{R} \times [0, \infty))$. Those that vanish near $t = 0$ test only the PDE in $t > 0$; those with $\varphi(\cdot, 0) \neq 0$ are what impose the *initial condition*, which enters solely through the term $\int u_0 \varphi(\cdot, 0)\, dx$ (an L^∞ function has no pointwise initial value to speak of).

4.2 Consistency with classical solutions

Proposition 4.2 (Consistency)**.** *If $u \in C^1(\mathbb{R}\times[0,\infty))$ satisfies* (5) *classically, it is a weak solution. Conversely, if a weak solution u happens to lie in $C^1(\mathbb{R}\times[0,\infty))$, it satisfies* (5) *classically.*

Proof sketch. The first claim is precisely the computation in (20), read in reverse. For the converse, reversing the integration by parts shows a C^1 weak solution satisfies $\int(u_t + f(u)_x)\varphi + \int(u(x,0) - u_0(x))\varphi(x,0) = 0$ for every $\varphi \in C_c^1$; taking φ supported away from $t = 0$ forces $u_t + f(u)_x = 0$ pointwise by the fundamental lemma of the calculus of variations, and taking $\varphi(\cdot, 0)$ arbitrary then forces $u(x, 0) = u_0(x)$. □

4.3 Piecewise smooth functions and the Rankine–Hugoniot condition

All discontinuous solutions in this paper have the following structure.

Definition 4.3 (Piecewise C^1 function)**.** Let $s_1, \dots, s_k : [0,\infty) \to \mathbb{R}$ be C^1 with $s_1(t) < \cdots < s_k(t)$ for $t > 0$ (the curves may start at a common point when $t = 0$). Put $s_0 \equiv -\infty$, $s_{k+1} \equiv +\infty$ and $\Omega_i = \{(x,t) :\ t > 0,\ s_i(t) < x < s_{i+1}(t)\}$ for $i = 0, \dots, k$. A bounded function u is *piecewise C^1 with respect to $s_1, \dots, s_k$* if on each Ω_i it is C^1 and u, u_x, u_t extend continuously to $\overline{\Omega_i} \cap \{t > 0\}$. Along $x = s_i(t)$ we write $u_\ell(t)$ and $u_r(t)$ for the limits taken from Ω_{i-1} and Ω_i respectively.

Theorem 4.4 (Rankine–Hugoniot condition)**.** *Let u be piecewise C^1 with respect to $s_1, \dots, s_k$, and a weak solution of* (5) *for some data $u_0 \in L^\infty$. Then $u_t + f(u)_x = 0$ classically in each Ω_i, and along each curve $x = s_i(t)$, $t > 0$,*

$$f(u_\ell(t)) - f(u_r(t)) = s_i'(t)\,\big(u_\ell(t) - u_r(t)\big). \tag{22}$$

In particular, wherever $u_\ell(t) \neq u_r(t)$,

$$s_i'(t) = \frac{f(u_\ell(t)) - f(u_r(t))}{u_\ell(t) - u_r(t)}. \tag{23}$$

Proof. Test functions φ supported in a single Ω_i give $\iint(u\varphi_t + f(u)\varphi_x) = 0$, so $u_t + f(u)_x = 0$ in the weak sense there; as u is C^1 on Ω_i, this holds classically.

Now fix $t_0 > 0$, an index i, and $\varphi \in C_c^1(\mathbb{R}\times(0,\infty))$ supported in a small ball around $(s_i(t_0), t_0)$ meeting no other curve. The ball is split by $x = s_i(t)$ into a left part in Ω_{i-1} and a right part in Ω_i. On each part, apply the divergence theorem to the C^1 vector field $\varphi\,(f(u), u)$ (components in the (x,t) order): since $u_t + f(u)_x = 0$ there,

$$\iint (u\varphi_t + f(u)\varphi_x)\,dx\,dt = \oint \varphi\,\big(f(u)\,n_x + u\,n_t\big)\,d\sigma,$$

the only boundary contribution being along the curve. The outward normal of the left part is proportional to $(1, -s_i')$ and that of the right part to $(-1, s_i')$, with $d\sigma\,\mathbf{n} = (1, -s_i')\,dt$ on the left. Adding the two parts and using (21),

$$0 = \int_0^\infty \Big[\big(f(u_\ell) - s_i'u_\ell\big) - \big(f(u_r) - s_i'u_r\big)\Big]\varphi(s_i(t), t)\,dt. \tag{24}$$

Since $\varphi(s_i(t), t)$ may be any C^1 function of t with small support, the bracket vanishes identically, which is (22). □

For Burgers' flux $f(u) = u^2/2$, (23) simplifies using $f(u_\ell) - f(u_r) = \frac{1}{2}(u_\ell - u_r)(u_\ell + u_r)$ to the familiar average-speed formula

$$s'(t) = \frac{u_\ell(t) + u_r(t)}{2}. \tag{25}$$

In words, a shock moves at the average of the two characteristic speeds (which equal the values of u) arriving from either side, consistent with the characteristic picture of Figure 2 below.

Theorem 4.4 gives necessary conditions. The next result is the converse, and is the tool used to verify every explicit solution in this paper.

Proposition 4.5 (Piecewise smooth weak solutions)**.** *Let $u_0 \in L^\infty(\mathbb{R})$ and let u be piecewise C^1 with respect to $s_1, \dots, s_k$ (Definition 4.3). Suppose*

(i) *$u_t + f(u)_x = 0$ classically in each Ω_i;*

(ii) *the jump relation* (22) *holds along each curve $x = s_i(t)$, $t > 0$ (automatic where u is continuous across the curve);*

(iii) (initial trace) $u(x,t) \to u_0(x)$ *as* $t \to 0^+$ *for every* $x \notin \{s_1(0), \dots, s_k(0)\}$.

Then u *is a weak solution of* (5) *with data* u_0.

Proof. Fix $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$ and $\varepsilon > 0$. On each $\Omega_i^\varepsilon = \Omega_i \cap \{t > \varepsilon\}$ (a Lipschitz domain, bounded after intersecting with $\operatorname{supp}\varphi$) the divergence theorem applied to $\varphi\,(f(u), u)$, together with (i), gives $\iint_{\Omega_i^\varepsilon}(u\varphi_t + f(u)\varphi_x)$ equal to the sum of boundary integrals over the curves and over the bottom edge $t = \varepsilon$. Adding over i, the contributions of the two sides of each curve combine, exactly as in (24), into $\int_\varepsilon^\infty [\dots]\varphi\,dt = 0$ by (ii). Only the bottom edge, with outward normal $(0,-1)$, survives:

$$\iint_{t>\varepsilon} \big(u\varphi_t + f(u)\varphi_x\big)\,dx\,dt = -\int_{\mathbb{R}} u(x,\varepsilon)\,\varphi(x,\varepsilon)\,dx.$$

Let $\varepsilon \to 0^+$. The left side tends to the integral over $t > 0$ (bounded integrand, compact support). On the right, $\varphi(\cdot,\varepsilon) \to \varphi(\cdot,0)$ uniformly and, by (iii), $u(\cdot,\varepsilon) \to u_0$ almost everywhere with $|u| \le \|u\|_\infty$, so dominated convergence gives $-\int u_0\varphi(\cdot,0)\,dx$. This is (21). □

Note that this argument does not require the curves to be well behaved at $t = 0$ (they may all emanate from one point, as for the rarefaction fans below), since it is run on $t > \varepsilon$ and the corner is handled by the limit.

5 Non-Uniqueness of Weak Solutions

Definition 4.1 was introduced to admit discontinuous solutions past the breakdown time of Proposition 3.1. This section shows that it admits *too many*: a single initial condition can have more than one weak solution, so the weak formulation alone does not determine the evolution.

5.1 The Riemann problem

Consider (5) with the specific piecewise-constant data

$$u_0(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases} \tag{26}$$

This is the simplest instance of a *Riemann problem*, and is studied in general in Section 7. The data are not C^1, so (5) is understood in the weak sense. Since u_0 is increasing across $x = 0$, the method of characteristics of Section 2 does not directly apply either: characteristics emanate from $x_0 < 0$ with speed 0 and from $x_0 > 0$ with speed 1, so no characteristic covers the region $0 < x < t$ at all. Two different weak solutions are constructed below, both consistent with this characteristic picture away from $0 < x < t$.

5.2 A shock solution

Define

$$u_A(x,t) = \begin{cases} 0, & x < st, \\ 1, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2} = \frac{0+1}{2} = \frac{1}{2}, \tag{27}$$

where s is chosen exactly to satisfy the Rankine–Hugoniot condition (25) with $u_\ell = 0$, $u_r = 1$.

Proposition 5.1. u_A *is a weak solution of* (5) *with data* (26).

Proof. Apply Proposition 4.5 with $k = 1$ and $s_1(t) = t/2$. (i) u_A is constant on each of the two regions, hence solves the PDE there. (ii) The jump from $u_\ell = 0$ to $u_r = 1$ moves at the Rankine–Hugoniot speed $s = 1/2$. (iii) For $x < 0$ we have $x < t/2$ so $u_A(x,t) = 0$, and for $x > 0$ we have $u_A(x,t) = 1$ once $t < 2x$; thus $u_A(\cdot,t) \to u_0$ at every $x \neq 0$. □

5.3 A rarefaction solution

Define instead

$$u_B(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t. \end{cases} \tag{28}$$

Proposition 5.2. u_B *is a weak solution of* (5) *with data* (26).

Proof. Apply Proposition 4.5 with $k = 2$, $s_1 \equiv 0$, $s_2(t) = t$ (two curves starting at the same point). (i) On the two outer regions u_B is constant. On $0 < x < t$, direct differentiation gives $(u_B)_t = -x/t^2$ and $(u_B)_x = 1/t$, so

$$(u_B)_t + u_B(u_B)_x = -\frac{x}{t^2} + \frac{x}{t} \cdot \frac{1}{t} = 0. \tag{29}$$

(ii) u_B is *continuous* across both curves $x = 0$ and $x = t$ (the one-sided limits are 0 at $x = 0$ and 1 at $x = t$), unlike u_A, so $u_\ell = u_r$ and (22) holds trivially, whatever the speed of the curve. (iii) For $x < 0$, $u_B = 0$; for $x > 0$, $u_B = 1$ once $t < x$. □

5.4 Two solutions, one initial condition

Corollary 5.3 (Non-uniqueness)**.** *The initial value problem* (5) *with data* (26) *has at least two distinct weak solutions, u_A and u_B.*

This is a genuine failure of uniqueness, not merely a technicality: u_A and u_B differ on the entire wedge $0 < x < t$, and both are constructed from, and consistent with, the same tools developed so far — u_A from the Rankine–Hugoniot condition of Section 4, and u_B from the self-similar construction justified in Section 7.1. Definition 4.1 alone gives no way to discard either one. In fact there are infinitely many:

Remark 5.4 (An infinite family). For every $\alpha \in \mathbb{R}$ let

$$u^\alpha(x,t) = \begin{cases} 0, & x < \frac{\alpha}{2}t, \\ \alpha, & \frac{\alpha}{2}t < x < \frac{1+\alpha}{2}t, \\ 1, & x > \frac{1+\alpha}{2}t. \end{cases}$$

The two jumps ($0 \to \alpha$ and $\alpha \to 1$) move at their Rankine–Hugoniot speeds $\alpha/2$ and $(1+\alpha)/2$, and $\alpha/2 < (1+\alpha)/2$, so Proposition 4.5 (with $k = 2$) shows each u^α is a weak solution with data (26). For $\alpha \notin \{0, 1\}$ these are distinct from u_A (and $u_A = u^0 = u^1$ as functions). Fans combined with shocks give further examples.

Remark 5.5. Physically, only u_B is correct: for the fluid interpretation of Section 1, increasing data $u_\ell < u_r$ should disperse smoothly (a rarefaction), not jump discontinuously (u_A is, in effect, a shock running *backwards* in time). Section 6 makes this precise with a condition that u_A and every u^α fail and u_B satisfies, restoring uniqueness.

6 The Entropy Condition

Corollary 5.3 showed that the weak formulation alone does not determine a unique solution. This section introduces additional admissibility conditions, motivated by the vanishing-viscosity limit made precise in Section 8, that select the physically correct weak solution. Two conditions appear, with different scopes: the *Lax condition* is a local test on individual jumps; the *Oleinik condition* is the global condition on which the uniqueness theory for general bounded weak solutions rests.

6.1 The Lax shock condition

Definition 6.1 (Lax shock condition)**.** Let u be a weak solution of a scalar conservation law $u_t + f(u)_x = 0$ with f convex, and suppose u has a jump discontinuity along a C^1 curve $x = s(t)$ with left and right limits $u_\ell(t)$, $u_r(t)$. The jump satisfies the *Lax shock condition* if

$$f'(u_\ell(t)) > s'(t) > f'(u_r(t)) \tag{30}$$

for all t.

For Burgers' equation, $f'(u) = u$, so (30) together with the Rankine–Hugoniot speed $s' = (u_\ell + u_r)/2$ reduces to

$$u_\ell > \frac{u_\ell + u_r}{2} > u_r \quad \Longleftrightarrow \quad u_\ell > u_r. \tag{31}$$

Remark 6.2. Condition (30) says the characteristics on either side of the shock must run *into* it as t increases, rather than out of it: information is being destroyed at the shock (consistent with physical irreversibility), not created. This is the precise sense in which a shock differs from a rarefaction, where characteristics instead fan *out* of a single point (Section 5.3). Note that the Lax condition constrains *jumps only*: it says nothing about the smooth part of a solution, nor about solutions that are not piecewise smooth. (A derivation of $u_\ell > u_r$ from viscosity alone, via the existence of viscous shock profiles, is given in Proposition 8.6.)

6.2 Entropy solutions: the Oleinik condition

For general bounded weak solutions (which need not be piecewise smooth) the right notion is a one-sided bound on the slope.

Definition 6.3 (Entropy solution)**.** A weak solution $u \in L^\infty$ of (5) (Burgers' flux $f(u) = u^2/2$) is an *entropy solution* if

$$u(x+z,t) - u(x,t) \;\le\; \frac{z}{t} \qquad \text{for a.e. } (x,t) \in \mathbb{R} \times (0,\infty) \text{ and all } z > 0. \tag{32}$$

Condition (32) is Oleinik's one-sided Lipschitz bound [12]: u may decrease abruptly (a shock) but can increase only gradually, at rate at most $1/t$. For a general uniformly convex flux the bound reads $u(x+z,t)-u(x,t) \le C(1+1/t)\,z$ for some constant C, and this is the form in which it is stated in the uniqueness theory of [5, §3.4]. Since $z/t \le C(1+1/t)z$ with $C = 1$, every solution satisfying (32) lies in that (larger) class, so the uniqueness theorem applies to ours. Conversely, for Burgers' equation the two classes coincide: by uniqueness, any solution in the larger class equals the Lax–Oleinik solution, which satisfies the sharp bound (32) (Remark 8.5). We therefore use the sharp form.

Lemma 6.4 (Oleinik implies Lax)**.** *Let u be a piecewise C^1 entropy solution. Then every jump of u satisfies the Lax condition* (30)*; for Burgers' equation, $u_\ell > u_r$ at every jump.*

Proof. Fix a curve $x = s(t)$ with $u_\ell(t) \neq u_r(t)$. Since u is continuous on each side of the curve, (32) may be applied (for a.e. t) with $x \uparrow s(t)$ and $x + z \downarrow s(t)$, so $z \to 0$. The left side tends to $u_r - u_\ell$ and the right side to 0; hence $u_r - u_\ell \le 0$, and since the limits differ, $u_\ell > u_r$. By continuity of u_ℓ, u_r in t this holds for every $t > 0$. □

Remark 6.5 (Scope). For a piecewise C^1 solution with locally finitely many jumps, one checks that (32) holds if and only if each jump satisfies $u_\ell > u_r$ *and* $u_x \le 1/t$ on the smooth pieces. So for the explicit solutions of this paper the two conditions agree, but the Lax condition by itself should not be taken as the definition of an entropy solution for arbitrary L^∞ weak solutions. In what follows the selection of the correct solution in each example is verified directly against (32).

6.3 Application to the Riemann problem of Section 5

Proposition 6.6. *For the Riemann data* (26) *($u_\ell = 0 < u_r = 1$), the rarefaction solution u_B of* (28) *is an entropy solution, while the shock solution u_A of* (27) *and every u^α of Remark 5.4 are not.*

Proof. u_B is entropy. Write $u_B(x,t) = \min\{\max\{x/t, 0\}, 1\}$, the composition of $x \mapsto x/t$ with the 1-Lipschitz function $y \mapsto \min\{\max\{y, 0\}, 1\}$. Hence $u_B(\cdot,t)$ is $(1/t)$-Lipschitz and $u_B(x+z,t) - u_B(x,t) \le z/t$, which is (32).

u_A is not. The jump $0 \to 1$ has $u_\ell < u_r$, violating (31), so by Lemma 6.4 u_A cannot be an entropy solution. Directly: for $x = t/2 - z/2$, $u_A(x+z,t) - u_A(x,t) = 1 > z/t$ as soon as $z < t$.

u^α is not. Each u^α contains an upward jump: if $\alpha > 0$ the jump $0 \to \alpha$ is upward, and if $\alpha \le 0$ the jump $\alpha \to 1$ is. (Both jumps cannot be downward, as that would need $0 > \alpha > 1$.) The same argument as for u_A applies. □

Proposition 6.6 recovers, as a formal criterion, the physical judgement made informally at the end of Section 5: among the weak solutions constructed there, the entropy condition selects exactly the rarefaction u_B.

6.4 Uniqueness

The entropy condition is not merely a rule of thumb for discarding individual bad solutions: it restores full well-posedness.

Theorem 6.7 (Oleinik; uniqueness of entropy solutions, [5, §3.4.3])**.** *For $u_0 \in L^\infty(\mathbb{R})$ and f smooth and uniformly convex (here $f(u) = u^2/2$), the initial value problem $u_t + f(u)_x = 0$, $u(\cdot,0) = u_0$, has at most one entropy solution in the sense of Definition 6.3, up to a set of measure zero. An entropy solution exists: the Lax–Oleinik solution [5, §3.4.2] is a weak solution satisfying* (32) *(Remark 8.5).*

Hypotheses and cited statement. Here "entropy solution" means the Oleinik class of Definition 6.3, not merely a weak solution whose jumps satisfy the Lax condition. The theorem as proved in [5, §3.4.3] is stated for a larger class (the constant $C(1 + 1/t)$ above), which contains ours; uniqueness there therefore implies uniqueness here.
The proof is beyond the scope of this paper. (Evans argues by duality, estimating the difference of two entropy solutions through a linear transport problem; the alternative formulation of Kruzhkov [8], via a family of entropy *inequalities*, gives an L^1-contraction and covers non-convex fluxes as well.) The point to record is the scope of the statement: uniqueness holds in the class defined by the Oleinik inequality (32), which for piecewise smooth solutions includes, but is not implied by, the Lax condition at jumps (Remark 6.5).

7 The Riemann Problem

Sections 5–6 treated the specific data $u_\ell = 0 < u_r = 1$. This section gives the complete entropy solution for arbitrary constant states $u_\ell, u_r \in \mathbb{R}$, i.e. the general *Riemann problem*

$$u_0(x) = \begin{cases} u_\ell, & x < 0, \\ u_r, & x > 0. \end{cases} \tag{33}$$

7.1 The rarefaction fan

The solution $u_B(x,t) = x/t$ constructed in Section 5.3 is a special case of a *self-similar* solution, i.e. one of the form $u(x,t) = g(x/t)$. Writing $\xi = x/t$ and substituting into the PDE gives

$$u_t + u\,u_x = -\frac{\xi}{t}g'(\xi) + \frac{g(\xi)}{t}g'(\xi) = \frac{g'(\xi)}{t}\big(g(\xi) - \xi\big) = 0, \tag{34}$$

so any self-similar solution satisfies either $g'(\xi) = 0$ (a constant state) or $g(\xi) = \xi$ (the fan $u = x/t$) on each interval of ξ. This is exactly the structure needed to connect two constant states, provided $u_\ell < u_r$: since characteristic speed equals u, the states u_ℓ and u_r correspond to characteristics of speed u_ℓ and u_r, and the fan $u = x/t$ smoothly interpolates the speeds in between whenever $u_\ell < u_r$.

7.2 The complete solution

Theorem 7.1 (Entropy solution of the Riemann problem)**.** *The unique entropy solution of* (5) *with data* (33) *is:*

(a) *If $u_\ell > u_r$ (shock case),*

$$u(x,t) = \begin{cases} u_\ell, & x < st, \\ u_r, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2}. \tag{35}$$

(b) *If $u_\ell \le u_r$ (rarefaction case),*

$$u(x,t) = \begin{cases} u_\ell, & x \le u_\ell t, \\ x/t, & u_\ell t < x < u_r t, \\ u_r, & x \ge u_r t. \end{cases} \tag{36}$$

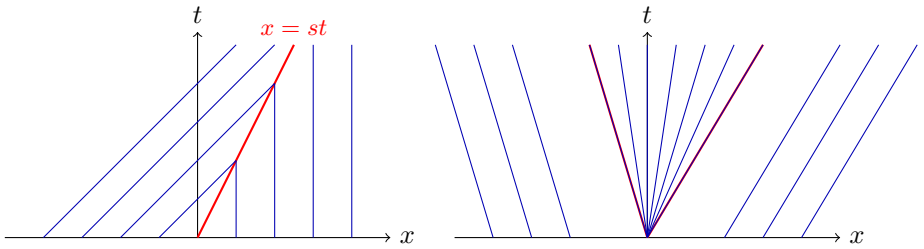
Proof. (a) Apply Proposition 4.5 with $k = 1$, $s_1(t) = st$: u is constant on each side (so (i) holds), the jump moves at the Rankine–Hugoniot speed $s = (u_\ell + u_r)/2$ (so (ii) holds), and for $x < 0$ (resp. $x > 0$) we have $x < st$ (resp. $x > st$) for small t, giving (iii). So u is a weak solution. It is an entropy solution: the only possible increments of $u(\cdot, t)$ are 0 and $u_r - u_\ell < 0$, so $u(x+z,t) - u(x,t) \le 0 \le z/t$, so (32) holds. Uniqueness is Theorem 6.7.

(b) The case $u_\ell = u_r$ is the trivial constant solution, so suppose $u_\ell < u_r$. Apply Proposition 4.5 with $k = 2$, $s_1(t) = u_\ell t$, $s_2(t) = u_r t$ (starting at a common point). (i) u is constant on the outer regions and equals x/t on the fan $u_\ell t < x < u_r t$, where it solves the PDE by Section 7.1. (ii) u is continuous across both curves (the one-sided limits agree, equalling u_ℓ and u_r respectively), so (22) holds trivially. (iii) For fixed $x < 0$ we have $x < u_\ell t$ for all small t, and for fixed $x > 0$, $x > u_r t$ for all small t, so $u(\cdot, t) \to u_0$ away from 0. Hence u is a weak solution. It is an entropy solution because $u(x,t) = \min\{\max\{x/t, u_\ell\}, u_r\}$ is $(1/t)$-Lipschitz in x, so (32) holds. Uniqueness is again Theorem 6.7. □

Taking $(u_\ell, u_r) = (0, 1)$ in case (b) recovers exactly u_B from Section 5.3, and taking $(u_\ell, u_r) = (1, 0)$ in case (a) gives the shock $s = 1/2$ analogous to u_A — now correctly identified as the entropy solution, since the roles of u_ℓ and u_r are reversed from Section 5 and the Lax condition (31) now holds.

8 Extensions

The core theory (Sections 2–7) is complete without this section. Subsections 8.1–8.3 reintroduce viscosity and connect the viscous and inviscid theories; Subsection 8.4 completes the worked example of Sections 2–3 using them; Subsections 8.5 and 8.6 are self-contained applications of the Riemann solver and of the entropy theory.



(a) shock: $u_\ell > u_r$ (b) rarefaction: $u_\ell < u_r$

Figure 2: Characteristic pictures for the two cases of Theorem 7.1. (a) With $u_\ell = 1$, $u_r = 0$ the shock has speed $s = 1/2$; characteristics from the left (speed $1 > s$) and from the right (speed $0 < s$) both run into the shock curve $x = st$. (b) Characteristics fan out from the origin, filling the gap left by the diverging characteristics from u_ℓ and u_r.

8.1 The Cole–Hopf transformation

(Due to Cole [3] and Hopf [7].)

This extension solves the viscous Burgers' equation (1) exactly by linearising it to the heat equation. The key observation is that, in one dimension, any u is the x-derivative of a potential, namely $-2\nu\,\partial_x$ of a logarithm.

Theorem 8.1 (Cole–Hopf transformation)**.** *Let u be a classical solution of $u_t + u\,u_x = \nu\,u_{xx}$ on $\mathbb{R}\times[0,T)$, and set*

$$\hat\psi(x,t) = -\frac{1}{2\nu}\int_0^x u(y,t)\,dy, \qquad c(t) = \int_0^t \Big(\frac{u(0,\tau)^2}{4\nu} - \frac{u_x(0,\tau)}{2}\Big)d\tau, \qquad \theta(x,t) = \exp\big(\hat\psi(x,t) + c(t)\big) > 0. \tag{37}$$

Then θ solves the heat equation $\theta_t = \nu\,\theta_{xx}$, and $u = -2\nu\,\theta_x/\theta$.

Proof. The identity $u = -2\nu\theta_x/\theta$ is immediate: $\theta_x/\theta = \hat\psi_x = -u/(2\nu)$, since c depends on t only. For the heat equation, let $w = \hat\psi_t - \nu\hat\psi_{xx} - \nu\hat\psi_x^2$. Using $\hat\psi_x = -u/(2\nu)$, $\hat\psi_{xx} = -u_x/(2\nu)$, $\hat\psi_{xt} = -u_t/(2\nu)$ and $\hat\psi_{xxx} = -u_{xx}/(2\nu)$,

$$w_x = \hat\psi_{xt} - \nu\hat\psi_{xxx} - 2\nu\hat\psi_x\hat\psi_{xx} = -\frac{1}{2\nu}\big(u_t + u\,u_x - \nu\,u_{xx}\big) = 0. \tag{38}$$

So w is a function of t alone, and may be evaluated at $x = 0$, where $\hat\psi(0,t) = 0$ gives $\hat\psi_t(0,t) = 0$; hence

$$w(t) = w(0,t) = \frac{u_x(0,t)}{2} - \frac{u(0,t)^2}{4\nu} =: C(t). \tag{39}$$

Put $\psi = \hat\psi + c$, so $c' = -C$ and $\psi_t - \nu\psi_{xx} - \nu\psi_x^2 = C + c' = 0$. Since $\theta = e^{\psi}$ gives $\theta_t = \theta\psi_t$ and $\theta_{xx} = \theta(\psi_{xx} + \psi_x^2)$, this is $\theta_t = \nu\theta_{xx}$. □

Remark 8.2. The correction $c(t)$ cannot be omitted in general. The naive function $e^{\hat\psi} = \exp\big(-\frac{1}{2\nu}\int_0^x u\,dy\big)$, normalised to equal 1 at $x = 0$, satisfies $\partial_t\theta = \nu\theta_{xx} + C(t)\theta$, which is the heat equation only if $C \equiv 0$. The factor $e^{c(t)}$ is independent of x, so it changes neither $u = -2\nu\theta_x/\theta$ nor the initial data ($c(0) = 0$).

8.2 Exact solution via the heat kernel

By Theorem 8.1, solving (1) reduces to solving the heat equation for θ with initial data

$$\theta_0(x) = \exp\left(-\frac{1}{2\nu}\int_0^x u_0(y)\,dy\right), \tag{40}$$

which, for bounded u_0, grows at most like $e^{\|u_0\|_\infty |x|/(2\nu)}$. The Cauchy problem for the heat equation has a unique solution in this growth class (see [5, §2.3.3]), given by convolution with the heat kernel,

$$\theta(x,t) = \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi\nu t}}\,\exp\left(-\frac{(x-y)^2}{4\nu t}\right)\theta_0(y)\,dy. \tag{41}$$

Substituting (40)–(41) into $u = -2\nu\,\theta_x/\theta$ and differentiating under the integral gives the explicit *Cole–Hopf formula*

$$u(x,t) = \frac{\displaystyle\int_{\mathbb{R}} \frac{x-y}{t}\,\exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}{\displaystyle\int_{\mathbb{R}} \exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}, \qquad G(x,y,t) = \int_0^y u_0(s)\,ds + \frac{(x-y)^2}{2t}. \tag{42}$$

Corollary 8.3 (Global smoothness; Hopf [7])**.** *For every bounded measurable u_0 and every $\nu > 0$, formula* (42) *defines a solution of* (1) *that is C^∞ for all $t > 0$; it is the unique bounded solution with data u_0.*

That (42) solves the equation follows by reversing the computation of Theorem 8.1: if $\theta > 0$ solves the heat equation, then $u = -2\nu\theta_x/\theta$ solves (1), and θ given by (41) is smooth and positive for $t > 0$. The statement that the initial data are attained and that the bounded solution is unique for merely measurable u_0 is due to Hopf [7]; it is cited, not reproved here.

Unlike the inviscid equation (Proposition 3.1), solutions of (1) stay smooth for all time, for any $\nu > 0$ however small: advection and viscous diffusion together do not by themselves produce singularities. This is a statement about Burgers' equation, not about Navier–Stokes: Burgers' scalar, one-dimensional structure (it obeys a maximum principle and has no pressure or vortex stretching) means that its regularisation mechanism is quite different from the open regularity question in three dimensions.

8.3 The vanishing viscosity limit

The Cole–Hopf solution also explains why the entropy conditions of Section 6 are the physically correct selection principle, rather than an arbitrary one.

Theorem 8.4 (Vanishing viscosity limit; Hopf, 1950)**.** *Let $u_0 \in L^\infty(\mathbb{R})$, let u^ν denote the solution of* (1) *with data u_0 given by* (42)*, and let u denote the entropy solution of* (2) *with the same data, in the sense of Definition 6.3 (for Riemann data this is Theorem 7.1). Then $u^\nu \to u$ pointwise almost everywhere, and in L^1_{loc}, as $\nu \to 0^+$ [7].*

The proof is a Laplace-method asymptotic analysis of (42) as $\nu \to 0$, and is not reproduced here. Its mechanism, however, is worth recording, because it is used again in Section 8.6.

Remark 8.5 (The Lax–Oleinik formula). As $\nu \to 0$ the weight $\exp(-G/2\nu)$ in (42) concentrates at the minimisers of $y \mapsto G(x, y, t)$. At every (x, t) where the minimiser $y^*(x, t)$ is unique, the limit is

$$u(x,t) = \frac{x - y^*(x,t)}{t}, \qquad y^*(x,t) = \arg\min_y \Big\{ V(y) + \frac{(x-y)^2}{2t} \Big\}, \quad V(y) = \int_0^y u_0, \tag{43}$$

the *Lax–Oleinik formula* [9, 12], [5, §3.4.2]. A jump occurs where two minimisers $y_1 < y_2$ tie, with $u_\ell = (x - y_1)/t > u_r = (x - y_2)/t$: the viscous limit produces only *downward* jumps, which is exactly the content of the Lax condition (31). More generally, $x \mapsto y^*(x, t)$ is nondecreasing (a standard property of the minimiser, [5, §3.4.2]), so for $z > 0$

$$u(x+z,t) - u(x,t) = \frac{z - \big(y^*(x+z,t) - y^*(x,t)\big)}{t} \;\le\; \frac{z}{t},$$

which is the sharp Oleinik bound (32). Together with the fact that (43) defines a weak solution [5, §3.4.2], this shows that the Lax–Oleinik solution is an entropy solution in the sense of Definition 6.3. Thus the entropy condition is not an ad hoc rule imposed to force uniqueness, but the criterion selected by the physically natural limit of vanishing viscosity — the same limit by which the inviscid equation (2) was justified as an idealisation of (1) in Section 1.

8.3.1 Viscous shock profiles and the Lax condition

The vanishing-viscosity picture can be made quite explicit at a single jump. It also shows directly that the Lax condition is not an arbitrary rule.

Proposition 8.6 (Viscous shock profiles)**.** *Let $u_\ell \neq u_r$ and $\nu > 0$. The viscous Burgers' equation* (1) *has a travelling-wave solution $u(x,t) = \phi(x - st)$, with $\phi \in C^2$ bounded and $\phi(\xi) \to u_\ell$, $\phi(\xi) \to u_r$ as $\xi \to -\infty, +\infty$ (and $\phi' \to 0$),* if and only if $u_\ell > u_r$. *In that case $s = (u_\ell + u_r)/2$ and, with $\Delta = u_\ell - u_r$,*

$$\phi(\xi) = s - \frac{\Delta}{2} \tanh\Big(\frac{\Delta\,\xi}{4\nu}\Big). \tag{44}$$

Moreover $\|\phi - (s - \frac{\Delta}{2}\operatorname{sgn}\xi)\|_{L^1(\mathbb{R})} = 4\ln 2\,\nu$, independent of s and Δ.

Proof. Substituting $u = \phi(x - st)$ gives $-s\phi' + \phi\phi' = \nu\phi''$, and integrating once from $-\infty$ (where $\phi' \to 0$),

$$\nu\phi' = \tfrac{1}{2}\phi^2 - s\phi - C, \qquad C = \tfrac{1}{2}u_\ell^2 - s u_\ell.$$

Letting $\xi \to +\infty$ shows the right-hand side must also vanish at u_r, so u_ℓ, u_r are the two roots of the quadratic: $s = (u_\ell + u_r)/2$ (the Rankine–Hugoniot speed) and

$$\nu\phi' = \tfrac{1}{2}(\phi - u_\ell)(\phi - u_r). \tag{45}$$

This is a first-order autonomous ODE, so ϕ cannot cross the equilibria u_ℓ, u_r and, to connect them, must lie strictly between them. If $u_\ell < u_r$, then for $\phi \in (u_\ell, u_r)$ the right side of (45) is *negative*, so ϕ decreases, yet it must rise from u_ℓ to u_r: no profile exists. If $u_\ell > u_r$ the right side is negative on (u_r, u_ℓ), consistent with a decrease from u_ℓ to u_r, and (45) is solved by (44). For the L^1 distance, by oddness,

$$\int_{\mathbb{R}} \tfrac{\Delta}{2}\Big|\tanh\tfrac{\Delta\xi}{4\nu} - \operatorname{sgn}\xi\Big|\,d\xi = \Delta\int_0^\infty \Big(1-\tanh\tfrac{\Delta\xi}{4\nu}\Big)d\xi = \Delta\cdot\frac{4\nu}{\Delta}\ln 2 = 4\ln 2\,\nu,$$

using $\int_0^\infty (1-\tanh s)\,ds = \ln 2$. □

Remark 8.7. (i) *The Lax condition from viscosity.* A jump with $u_\ell < u_r$ is not the limit of viscous travelling waves, whereas every jump with $u_\ell > u_r$ is, with thickness $\sim \nu/\Delta \to 0$. This derives the admissibility condition (31) from the vanishing-viscosity principle directly, and explains why u_A of Section 5 is rejected: it contains an upward jump, which no vanishing-viscosity limit can produce. (ii) *Galilean invariance.* Equation (1) is invariant under $u \mapsto u + c$, $x \mapsto x + ct$, which maps a profile of speed s to one of speed $s - c$ and leaves L^1 distances between u-values unchanged. This is why the constant $4\ln 2\,\nu$ does not depend on the speed s: for the stationary shock of strength a ($s = 0$, $\Delta = 2a$) it is the profile $-a\tanh(ax/2\nu)$ used in Section 8.4, and for the shock $1 \to 0$ of Section 8.5.5 ($s = \frac{1}{2}$, $\Delta = 1$) it is the same profile boosted by $\frac{1}{2}$.

8.4 The worked example completed: $u_0 = -\arctan x$

Sections 2.5 and 3.4 followed $u_0(x) = -\arctan x$ up to its breakdown at $(x^*, t^*) = (0, 1)$. The Lax–Oleinik formula (43) lets us continue the entropy solution explicitly. The results of this subsection are worked computations (with the proofs given in outline), not new theorems; the numerical parts are produced by `code/arctan_example.py`.

8.4.1 Exact entropy solution after breakdown

Here $V(y) = \int_0^y u_0 = -\big(y\arctan y - \frac{1}{2}\ln(1+y^2)\big)$, so the critical points of $y \mapsto G(y) = V(y) + (x-y)^2/(2t)$ satisfy

$$G'(y) = -\arctan y + \frac{y-x}{t} = 0 \iff x = y - t\arctan y = \Phi_t(y), \tag{46}$$

i.e. they are exactly the foot points $x_0 = y$ of the characteristics through (x, t). For $t > 1$ there are three of them when $|x|$ is small.

Proposition 8.8 (Entropy solution for $u_0 = -\arctan x$)**.** *For $t < 1$ the entropy solution is the classical solution* (10) *(at $t = 1$ it is continuous, with a vertical tangent at $x = 0$). For $t > 1$ let $y^*(t) > 0$ be the unique positive root of* $\arctan y = y/t$. *Then the entropy solution is odd in x, smooth for $x \neq 0$, and for $x > 0$*

$$u(x,t) = -\arctan y, \qquad \text{where } y \geq y^*(t) \text{ is the unique solution of } y - t\arctan y = x, \tag{47}$$

with a single stationary *shock at $x = 0$ of strength*

$$u_\ell(t) = -u_r(t) = \arctan y^*(t) \in (0, \pi/2). \tag{48}$$

Moreover $u_\ell(t) = \sqrt{3(t-1)}\,(1+o(1))$ as $t \downarrow 1$, and $u_\ell(t) \to \pi/2$ as $t \to \infty$.

Proof (outline). For $t > 1$, $G(\cdot)$ at $x = 0$ has $G'(y) = y/t - \arctan y$, so $y = 0$ is a critical point with $G''(0) = (1-t)/t < 0$ (a local *maximum*). Since $\arctan y/y$ decreases from 1 to 0 on $(0, \infty)$, there is exactly one positive root y^* of $\arctan y = y/t$, and by oddness of G' the minimisers at $x = 0$ are the two points $\pm y^*$: a tie, hence a jump at $x = 0$, with $u_\ell = (0 - (-y^*))/t = y^*/t = \arctan y^*$ and $u_r = -u_\ell$. From $\arctan y > y/(1+y^2)$ for $y > 0$ we get $1 + y^{*2} > t$, so $\Phi_t'(y) = 1 - t/(1+y^2) > 0$ for $y \geq y^*$: on $[y^*, \infty)$ the map Φ_t is strictly increasing from 0 to ∞. For $x > 0$, every minimiser is $\geq y^*$ because the minimiser is nondecreasing in x [5, §3.4.2]; so the minimiser is the unique root of (46) in $[y^*, \infty)$, which gives (47) (using $u = (x-y)/t = -\arctan y$), and in particular there is no other jump. Oddness gives $x < 0$. For the asymptotics, write $\arctan y^* = y^*/t$ and $y^* \approx \sqrt{3(1-1/t)}$ from $\arctan y = y - y^3/3 + O(y^5)$ near $t = 1$; and for $t \geq 2$ the inequality $1 + y^{*2} > t$ forces $y^* > 1$, so $y^* = t\arctan y^* \geq t\pi/4 \to \infty$ and $\arctan y^* \to \pi/2$. □

Thus the shock is born at $(0, 1)$ with zero strength, grows like $\sqrt{3(t-1)}$, and saturates at $\pi/2$ — the full jump between the limits $u_0(-\infty) = \pi/2$ and $u_0(+\infty) = -\pi/2$ of the data, so $u(\cdot, t) \to -\frac{\pi}{2}\operatorname{sgn} x$. This replaces the symmetry remark of Section 3.4 by an explicit shock strength (Figure 3, left). Numerically, the parametrisation (47) agrees with a brute-force minimisation of G to within the grid resolution ($\lesssim 5 \times 10^{-4}$ for $t \in \{0.5, 1, 2, 5\}$), and $u_\ell/\sqrt{3(t-1)} = 0.9999,\ 0.9994,\ 0.994,\ 0.945$ for $t - 1 = 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}$.

8.4.2 Vanishing viscosity for this example: a prediction and a check

Theorem 8.4 asserts that the Cole–Hopf solutions u^ν converge to this entropy solution. Here we can predict the size of the error, not just its vanishing.

Prediction. Near the shock, u^ν should look like a stationary viscous shock. By Proposition 8.6 (with $s = 0$ and $\Delta = 2a$) such a shock is $-a\tanh(ax/2\nu)$, and its L^1 distance from the sharp shock $-a\operatorname{sgn} x$ is $4\ln 2\,\nu \approx 2.77\,\nu$, independent of the strength a. So if u^ν is locally a stationary shock of slowly varying strength $a = u_\ell(t)$, the layer contributes $\approx 4\ln 2\,\nu$ to $\|u^\nu - u\|_{L^1}$, and the rest of the error comes from the smooth part, as an $O(\nu)$ viscous correction.

Check. We evaluated (42) with the above V by quadrature (log-sum-exp stabilisation of the weights $e^{-G/2\nu}$) and compared with Proposition 8.8 at $t = 3$ (Figure 3, centre). Table 1 gives the $L^1(-6,6)$ error: it is first order in ν and tends to $\approx 3.11\,\nu$. Splitting the integral at $|x| = 0.05$ gives a layer contribution of $2.78\,\nu$ (for $\nu = 0.003$, 0.001, 0.0003) against the predicted $2.77\,\nu$, within 0.5%, and a smooth-part contribution of $\approx 0.33\,\nu$.

ν	0.1	0.03	0.01	0.003	0.001	0.0003	0.0001
$\|u^\nu - u\|_{L^1(-6,6)}$	3.16×10^{-1}	9.36×10^{-2}	3.11×10^{-2}	9.34×10^{-3}	3.11×10^{-3}	9.33×10^{-4}	3.10×10^{-4}
error$/\nu$	3.156	3.122	3.114	3.112	3.111	3.110	3.104
order (vs. previous ν)	—	1.01	1.00	1.00	1.00	1.00	1.00

Table 1: L^1 error of the Cole–Hopf solution against the entropy solution at $t = 3$.

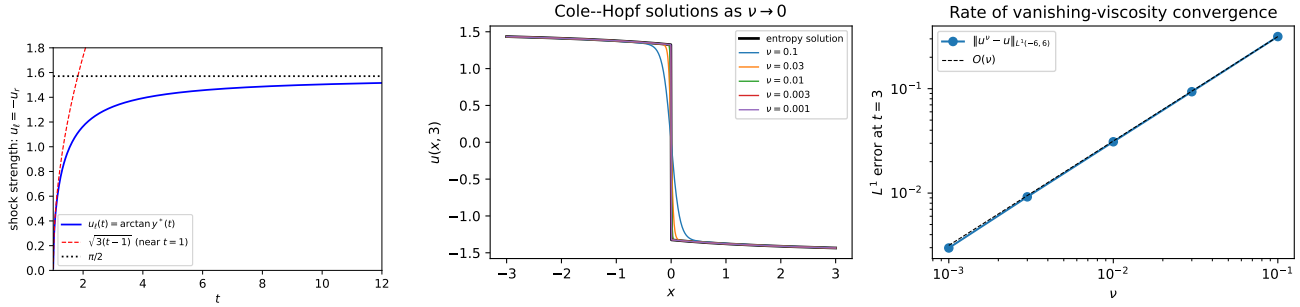


Figure 3: Left: shock strength $u_\ell(t) = \arctan y^*(t)$ for $u_0 = -\arctan x$, with the small-$(t-1)$ asymptote and the limit $\pi/2$. Centre: Cole–Hopf solutions at $t = 3$ for decreasing ν against the entropy solution (black). Right: L^1 error against ν (log–log).

Is the layer error independent of the strength? The prediction says the layer contribution does not depend on a, whereas the data of Table 1 all have $a = u_\ell(3) \approx 1.32$. We repeated the measurement at other times, i.e. other strengths (`code/strength_independence.py`, $\nu = 3\times10^{-4}$), computing the ratio of the error inside $|x| < \delta$ to $4\ln 2\,\nu$ (prediction: 1).

t	strength a	outer slope $u_x(0^+)$	ratio, $\delta = 0.05$	ratio, $\delta = 0.005$
1.1	0.518	−4.46	1.464	1.087
1.5	0.967	−0.62	1.054	1.006
3	1.324	−0.073	1.004	1.0004
10	1.504	−0.005	1.000	0.9999

Table 2: Layer error divided by $4\ln 2\,\nu$ for several shock strengths.

The ratio tends to 1 as the window shrinks, at every strength tested. With the wider window it deviates when the outer solution is steep beside the shock (young shocks, t close to 1, large $|u_x(0^+)|$): then the smooth part's $O(\nu)$ correction, which lies inside the window, is large, and at $t = 1.1$ the ratio at $\delta = 0.005$ is still 9% above 1 and not converged. So the prediction is confirmed where the shock is established (its strength varies by a factor of 3 across the table), and the early deviations are explained by contamination from the outer solution, not by a failure of the layer picture; this explanation is supported by the trend in δ but not proved.

These are numerical observations for one data set, not theorems; for general bounded data one only expects the weaker convergence of Theorem 8.4. The prediction identifies the layer part of the error exactly, but the $0.33\,\nu$ smooth-part constant is only measured, not derived.

8.5 A numerical scheme: Godunov's method

This extension outlines Godunov's method [6], a numerical scheme for the conservation law $u_t + f(u)_x = 0$ that uses the exact Riemann solution of Theorem 7.1 as its building block, and states its convergence to the entropy solution.

8.5.1 Finite volume discretisation

Divide $\mathbb{R}$ into cells $[x_{j-1/2}, x_{j+1/2}]$ of width Δx, with cell averages

$$U_j^n \approx \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t^n)\, dx, \tag{49}$$

at times $t^n = n\Delta t$. Integrating $u_t + f(u)_x = 0$ over a cell and over $[t^n, t^{n+1}]$ gives the exact identity

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{\Delta x} \left(\widehat{f}_{j+1/2} - \widehat{f}_{j-1/2} \right), \tag{50}$$

where $\widehat{f}_{j+1/2}$ is the (time-averaged) flux through the interface $x_{j+1/2}$, which must be approximated by a *numerical flux function* $\widehat{f}_{j+1/2} = \widehat{f}(U_j^n, U_{j+1}^n)$.

8.5.2 The Godunov flux

Godunov's method defines $\widehat{f}$ by solving, at each interface, the Riemann problem with left and right states U_j^n and U_{j+1}^n *exactly*, using Theorem 7.1, and evaluating the resulting self-similar solution along $x/t = 0$:

$$\widehat{f}_{j+1/2} = f\big(u^R(0;\, U_j^n, U_{j+1}^n)\big), \tag{51}$$

where $u^R(\xi; u_\ell, u_r)$ denotes the value at self-similar coordinate $\xi = x/t$ of the entropy solution from Theorem 7.1 with data (u_ℓ, u_r). Concretely, by (35)–(36),

$$u^R(0; u_\ell, u_r) = \begin{cases} u_\ell, & u_\ell > u_r,\ s = \frac{u_\ell + u_r}{2} > 0, \\ u_r, & u_\ell > u_r,\ s < 0, \\ u_\ell, & u_\ell \le u_r,\ u_\ell \ge 0, \\ u_r, & u_\ell \le u_r,\ u_r \le 0, \\ 0, & u_\ell < 0 < u_r, \end{cases} \tag{52}$$

the last case corresponding to $\xi = 0$ lying inside the rarefaction fan of (36), i.e. $u^R(0) = (f')^{-1}(0) = 0$ for Burgers' flux. (In the shock case with $s = 0$, i.e. $u_\ell = -u_r$, the value of $u^R(0)$ is ambiguous, but $f(u_\ell) = f(u_r)$ so the flux (51) is the same either way.)

8.5.3 Stability and convergence

Proposition 8.9 (CFL condition)**.** *The Godunov scheme is stable provided the time step satisfies*

$$\Delta t \le \frac{\Delta x}{\max_j |f'(U_j^n)|}. \tag{53}$$

Condition (53) (the Courant–Friedrichs–Lewy condition) ensures that information does not propagate across more than one cell per time step, consistent with the finite propagation speed of characteristics established in Section 2. Two distinct requirements are worth separating (see [10, Ch. 4]). The flux (51) is the exact flux through $x_{j+1/2}$ as long as the state at that interface stays constant during the step, i.e. no wave from a neighbouring interface reaches it; this holds for CFL number ≤ 1, which is (53), and is all that stability and monotonicity of the scheme need. The "reconstruct–evolve–average" picture, in which the exact solution in each cell is the superposition of the Riemann solutions from its two interfaces, requires in addition that these two sets of waves do not interact inside the cell, i.e. CFL number $\le \frac{1}{2}$. The value 0.45 used below satisfies both.

Theorem 8.10 (Convergence; [4], [10, Ch. 12])**.** *Under* (53)*, the Godunov approximations converge, as $\Delta x, \Delta t \to 0$, to the unique entropy solution of Theorem 6.7.*

Remark 8.11. This gives a second, independent route to the solutions constructed by hand in Sections 5–7: an implementation of (51) applied to the Riemann data (26) should reproduce the shock speed $s = 1/2$ and the linear profile $u = x/t$ inside the rarefaction fan, providing a computational check on the exact formulas derived analytically above.

8.5.4 Numerical results

Setup. The scheme of Sections 8.5.1–8.5.2 was implemented (Python/NumPy; script `code/godunov.py`) on $x \in [-2, 2]$ with N cells, initial cell averages of the Riemann data (the discontinuity lies on a cell interface), zero-gradient (outflow) boundary conditions, time step $\Delta t = 0.45\,\Delta x / \max_j |U_j^n|$ (CFL number 0.45 in (53), last step shortened to land on $t = 1$), and was run to $t = 1$ on the rarefaction data $(u_\ell, u_r) = (0, 1)$ of Section 5 and the shock data $(u_\ell, u_r) = (1, 0)$. The error is the L^1 error $\|U - u\|_{L^1(-2,2)} = \int |U - u|\, dx$ against the exact entropy solution of Theorem 7.1, with the exact solution integrated over each cell (so the cell containing the shock is not mis-scored). Two sanity checks were passed: the case-based flux of (51) agrees with the closed form $\min_{[u_\ell,u_r]} f$ (if $u_\ell \le u_r$), $\max_{[u_r,u_\ell]} f$ (otherwise) on 10^5 random state pairs, and total mass equals the initial mass (2) plus the net boundary flux, namely -0.5 for the rarefaction problem (final mass 1.5) and $+0.5$ for the shock problem (final mass 2.5), to ten digits. Figure 4 compares the numerical solution with $N = 400$ against the exact solution.

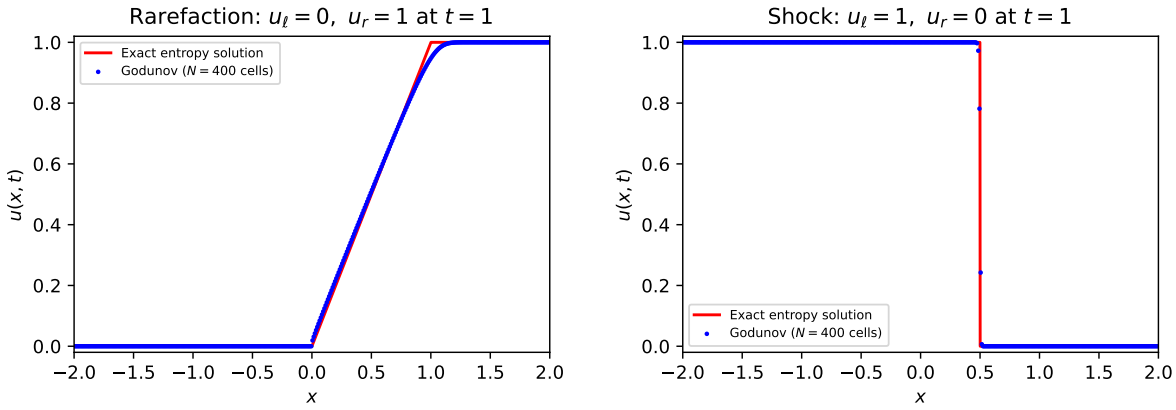


Figure 4: Godunov scheme (blue dots, $N = 400$ cells) against the exact entropy solution (red) of Theorem 7.1, at $t = 1$. Left: rarefaction data $(u_\ell, u_r) = (0, 1)$; the scheme reproduces the linear fan $u = x/t$, with smearing at the two corners (visibly larger at the upper corner $x = 1$) typical of a first-order scheme. Right: shock data $(u_\ell, u_r) = (1, 0)$; the discontinuity is located at the Rankine–Hugoniot position $x = st = 0.5$ (eq. (25)) and is spread over about three cells, with no overshoot or oscillation, as expected of a monotone scheme.

Convergence. Table 3 and Figure 5 report the L^1 error for $N = 200, 400, \ldots, 6400$ (the choice of these resolutions is explained below).

N	Δx	L^1 (rarefaction)	order	L^1 (shock)	order
200	2.0×10^{-2}	3.05×10^{-2}	—	9.91×10^{-3}	—
400	1.0×10^{-2}	1.82×10^{-2}	0.74	4.98×10^{-3}	0.99
800	5.0×10^{-3}	1.07×10^{-2}	0.77	2.50×10^{-3}	0.99
1600	2.5×10^{-3}	6.14×10^{-3}	0.80	1.24×10^{-3}	1.01
3200	1.25×10^{-3}	3.48×10^{-3}	0.82	6.23×10^{-4}	0.99
6400	6.25×10^{-4}	1.95×10^{-3}	0.84	3.13×10^{-4}	0.99

Table 3: L^1 errors of the Godunov scheme at $t = 1$ and observed orders $\log_2(e_{N/2}/e_N)$ between consecutive rows. All N are multiples of 8, so the shock lies on a cell interface (see the text).

Shock: dependence on grid alignment. The shock at $x = st = 0.5$ lies on a cell interface exactly when $0.5/\Delta x = N/8$ is an integer, i.e. when $8 \mid N$. For such N the shock error is $0.50\,\Delta x$ to two digits (values 0.495–0.501; the observed order is ≈ 1.0, as in Table 3). For other N it depends on where the shock sits within a cell: for $N = 97, \ldots, 103$ the error ranges from $0.60\,\Delta x$ to $0.77\,\Delta x$ (e.g. $0.655\,\Delta x$ at $N = 100$, $0.746\,\Delta x$ at $N = 97$), against $0.50\,\Delta x$ at $N = 96$ and $N = 104$. Observed orders between a misaligned and an aligned resolution are therefore unreliable: $N = 100 \to 200$ would give an order of 1.40 for the shock. This is why the table uses $N = 200 \cdot 2^k$. In every case the error is proportional to Δx, consistent with the discontinuity being resolved over a fixed number of cells.

Rarefaction. Here the error also decreases monotonically, but the observed order is ≈ 0.7–0.8 and is still slowly increasing $(0.74, 0.77, 0.80, 0.82, 0.84)$, so these experiments do not by themselves establish a first-order rate. The larger error is attributable to the two corners of the fan, where the exact solution is only Lipschitz and the scheme's numerical diffusion smears it; we do not claim more than the data show. (Theorem 8.10 asserts convergence, not a

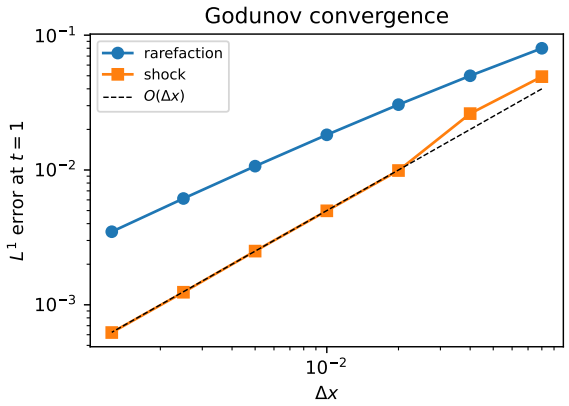


Figure 5: L^1 error at $t = 1$ against Δx (log–log), with an $O(\Delta x)$ reference line.

rate.) In both cases the computed shock position and fan slope agree with the analytic values $s = 1/2$ and $u = x/t$ of Section 7, which is the check proposed in Remark 8.11.

8.5.5 Numerical diffusion as vanishing viscosity

This subsection tests a heuristic that connects the scheme to the guiding question: a monotone scheme converges to the *entropy* solution because its numerical diffusion acts as a vanishing viscosity (the mechanism of Section 8.3). It is a consistency check, not a proof; the rigorous statement is Theorem 8.10.

In the shock problem all states lie in $[0, 1]$ and $s = \frac{1}{2} > 0$, so the Godunov flux is the upwind flux and the scheme reads $U_j^{n+1} = U_j^n - \lambda(f(U_j^n) - f(U_{j-1}^n))$ with $\lambda = \Delta t/\Delta x$. A Taylor expansion gives the modified equation

$$u_t + u\,u_x = \frac{\Delta x}{2}\big(u(1-\lambda u)\,u_x\big)_x + O(\Delta x^2), \tag{54}$$

a viscous Burgers' equation with state-dependent viscosity $\nu(u) = \frac{\Delta x}{2}u(1-\lambda u) \geq 0$ exactly when the CFL condition (53) holds. Its travelling shock from 1 to 0, with speed $\frac{1}{2}$, satisfies $\nu(u)\,u_\xi = \frac{1}{2}u(u-1)$ in $\xi = x - t/2$, i.e. in units of Δx

$$\frac{du}{d\xi} = \frac{u-1}{1-\lambda u}, \qquad \xi(u) = \xi_0 + (1-\lambda)\ln(1-u) - \lambda u. \tag{55}$$

The profile decays exponentially to 1 as $\xi \to -\infty$ and reaches 0 at a finite ξ_0. Conservation of mass (the scheme is conservative) fixes $\xi_0 = 1 - \lambda/2$, and the L^1 distance from the sharp shock is

$$\frac{\|u - u_{\text{exact}}\|_{L^1}}{\Delta x} = 2\int_0^{u_0} \frac{u(1-\lambda u)}{1-u}\,du, \qquad (1-\lambda)\ln(1-u_0) - \lambda u_0 = -\Big(1 - \frac{\lambda}{2}\Big). \tag{56}$$

Table 4 compares this with the measured shock error on aligned grids (here λ equals the CFL number, since $\max|U| = 1$); the code is `code/numerical_viscosity.py`.

CFL $= \lambda$	measured ($N = 1600$)	predicted	difference
0.10	0.627	0.686	+9.4%
0.20	0.589	0.635	+8.0%
0.30	0.553	0.586	+5.8%
0.45	0.496	0.511	+3.1%
0.60	0.448	0.438	−2.4%
0.80	0.352	0.342	−3.0%
0.95	0.300	0.272	−9.3%

Table 4: Shock L^1 error divided by Δx: measured Godunov (grids with $8 \mid N$) against the modified-equation prediction.

The prediction is within 3% at the CFL number 0.45 used above and within 10% across the whole range, and it reproduces the observed decrease of the error with λ. At CFL 0.95 the measured value is still resolution dependent (0.334 at $N = 400$, 0.300 at $N = 1600$), so that row is not asymptotic. Equating the error to the viscous-shock value

$4\ln 2\,\nu_{\text{eff}}$ of Proposition 8.6 (valid for any shock speed, here $s=\frac12$, $\Delta=1$, by Galilean invariance) defines an effective viscosity: the measured $0.50\,\Delta x$ gives $\nu_{\text{eff}}\approx 0.180\,\Delta x$, the modified-equation profile gives $0.184\,\Delta x$, and the cruder estimate $\frac{\Delta x}{2}u(1-\lambda u)$ at $u=\frac12$ gives $0.194\,\Delta x$. Since $\nu_{\text{eff}}=O(\Delta x)\to 0$, in the shock layer the scheme behaves like a vanishing viscosity, which is the numerical counterpart of Theorem 8.4. This agreement should not be over-read: the numerical shock occupies only a few cells, and (54) neglects the $O(\Delta x^2)$ terms.

8.6 Long-time behaviour: the N-wave

This final extension describes the universal shape that entropy solutions with compactly supported bounded initial data approach as $t\to\infty$, regardless of the detailed shape of u_0. The point to keep in mind is that the two constants specifying this shape are *not* the total positive and negative masses of u_0: positive and negative parts of the data can annihilate each other across shocks.

8.6.1 An exact self-similar solution

Proposition 8.12 (The N-wave)**.** *Fix constants $P,Q>0$. The function*

$$\mathcal{N}(x,t)=\begin{cases}0, & x\le -\sqrt{2Qt},\\ x/t, & -\sqrt{2Qt}<x<\sqrt{2Pt},\\ 0, & x\ge\sqrt{2Pt},\end{cases}\qquad t>0, \tag{57}$$

is an entropy solution of $u_t+u\,u_x=0$ on $\mathbb{R}\times[t_0,\infty)$ for every $t_0>0$ (with data $\mathcal{N}(\cdot,t_0)$). Moreover $\int\mathcal{N}^+(x,t)\,dx=P$, $\int\mathcal{N}^-(x,t)\,dx=Q$ for all $t>0$.

Proof. Apply Proposition 4.5 on $t\ge t_0$ with the two curves $x_-(t)=-\sqrt{2Qt}$ and $x_+(t)=\sqrt{2Pt}$, which are C^1 there. (i) On the middle interval $\mathcal{N}=x/t$ is the self-similar fan of Section 7.1, which solves the PDE classically; the outer regions are constant. (ii) At the right edge the one-sided limits are $u_\ell=x_+(t)/t=\sqrt{2P/t}$ and $u_r=0$; since $\dot x_+(t)=\sqrt{P/(2t)}=\frac12\big(\sqrt{2P/t}+0\big)$, this is exactly the Rankine–Hugoniot speed (25). At the left edge $u_\ell=0$, $u_r=-\sqrt{2Q/t}$, and $\dot x_-(t)=-\sqrt{Q/(2t)}=\frac12\big(0+(-\sqrt{2Q/t})\big)$, again matching (25). (iii) The initial trace at t_0 is $\mathcal{N}(\cdot,t_0)$ by definition. Hence $\mathcal{N}$ is a weak solution there.

For the entropy condition (32): both jumps are downward ($u_\ell>u_r$), and inside the middle interval $N_x=1/t$. If x and $x+z$ both lie in the middle interval, $\mathcal{N}(x+z,t)-N(x,t)=z/t$; if x lies left of it, $\mathcal{N}(x+z,t)\le z/t$ because $x<0$. The only remaining case with a possible increase is $x<0$ inside the middle interval and $x+z$ beyond the right edge; then $\mathcal{N}(x+z,t)-N(x,t)=-x/t=|x|/t\le z/t$, because $x+z>0$ forces $|x|<z$. So (32) holds. Finally $\int_0^{\sqrt{2Pt}}(x/t)\,dx=P$ and $\int_{-\sqrt{2Qt}}^0(-x/t)\,dx=Q$. □

The profile (57) is called an *N-wave*: it rises linearly through the origin and is bounded by a shock on each side, so that its graph traces the shape of the letter N (reversed), decaying in amplitude like $t^{-1/2}$ and spreading in width like $t^{1/2}$ as $t\to\infty$. (If $P=0$ or $Q=0$ the corresponding shock is absent and the profile is a single triangle ending in one shock; $P=Q=0$ gives $\mathcal{N}\equiv 0$.) The function $\mathcal{N}$ is unbounded as $t\to 0^+$, so it is not a weak solution on $t>0$ in the sense of Definition 4.1; as $t\to 0^+$ it converges, in the sense of distributions, to $(P-Q)\delta_0$: both edges collapse to the origin, a state concentrated entirely at a point.

8.6.2 Shock merging and cancellation

Two mechanisms are visible as general solutions evolve towards the N-wave; both are illustrations of why such a limit is plausible, and neither, by itself, proves the convergence theorem of Section 8.6.3. The first is that colliding shocks merge into one, governed by exactly the Riemann-problem machinery already developed.

Lemma 8.13 (Shock merging)**.** *Let $u_1>u_2>u_3$, and consider initial data equal to u_1 for $x<a$, u_2 for $a<x<b$, and u_3 for $x>b$. For small $t>0$ the entropy solution consists of two shocks moving at speeds $s_{12}=(u_1+u_2)/2$ and $s_{23}=(u_2+u_3)/2$; since $s_{12}>s_{23}$, they collide at the finite time $t^\times=2(b-a)/(u_1-u_3)$, after which the entropy solution is the single shock connecting u_1 to u_3 at speed $s_{13}=(u_1+u_3)/2$.*

Proof sketch. Each jump individually satisfies the Lax condition since $u_1>u_2>u_3$, so Theorem 7.1(a) gives the two shock speeds above. Since $s_{12}-s_{23}=\frac12(u_1-u_3)>0$, the shock positions $a+s_{12}t$ and $b+s_{23}t$ meet at the stated finite $t^\times$. At that instant the middle state has shrunk to zero width, leaving the two-state data u_1,u_3 on either side of the collision point; since $u_1>u_3$, Theorem 7.1(a) applied afresh from time $t^\times$ gives the merged shock at speed s_{13}. □

Remark 8.14 (Scope of the lemma). Lemma 8.13 covers only shock–shock collisions. In general a shock also absorbs rarefaction fans, and a shock separating a positive state (on its left) from a negative one (on its right) cancels positive against negative mass. The following example shows this cancellation concretely and why it matters.

Example 8.15 (Cancellation)**.** Let $u_0 = +1$ on $(0,1)$, $u_0 = -1$ on $(1,2)$, and $u_0 = 0$ elsewhere. For $0 < t < 1$ the entropy solution is

$$u = \begin{cases} 0, & x < 0, \\ x/t, & 0 < x < t, \\ 1, & t < x < 1, \\ -1, & 1 < x < 2 - t, \\ (x-2)/t, & 2 - t < x < 2, \\ 0, & x > 2, \end{cases}$$

(two fans, and a stationary shock at $x = 1$, since $u_\ell = 1 > u_r = -1$ and $(u_\ell + u_r)/2 = 0$), as one checks with Proposition 4.5. At $t = 1$ the fans reach the shock, and for $t \geq 1$

$$u(x,t) = \begin{cases} x/t, & 0 < x < 1, \\ (x-2)/t, & 1 < x < 2, \\ 0, & \text{otherwise}, \end{cases}$$

with the stationary shock at $x = 1$ ($u_\ell = 1/t$, $u_r = -1/t$) and continuity elsewhere; it satisfies (32) because its only jump is downward and $u_x = 1/t$ elsewhere. Hence for $t \geq 1$, $\|u(\cdot,t)\|_{L^1} = \frac{1}{2t} + \frac{1}{2t} = \frac{1}{t} \to 0$, even though $\int u_0^+ = \int u_0^- = 1$.

8.6.3 Convergence of general solutions to the N-wave

The correct constants are read off from the primitive of u_0, not from its positive and negative parts.
The following is a *cited* theorem: it is stated here without proof, and the reader should consult [9] and [5, §3.4.5] for the precise hypotheses, norm and proof.

Theorem 8.16 (Asymptotic N-wave decay; Lax [9], see also [5, §3.4.5])**.** *Let $u_0 \in L^\infty(\mathbb{R})$ have compact support, and set*

$$V(y) = \int_{-\infty}^{y} u_0(x)\,dx, \qquad M = \int_{\mathbb{R}} u_0\,dx, \qquad Q = -\min_y V(y), \qquad P = M - \min_y V(y). \tag{58}$$

Then $P, Q \geq 0$, $P - Q = M$, and the entropy solution u of (5) satisfies

$$\|u(\cdot,t) - \mathcal{N}(\cdot,t)\|_{L^1(\mathbb{R})} \to 0 \qquad \text{as } t \to \infty, \tag{59}$$

where $\mathcal{N}$ is the N-wave (57) with these constants P, Q.

(The primitive V here is based at $-\infty$, whereas in Remark 8.5 it is based at 0; the two differ by a constant, which changes neither the minimisers of $V(y) + (x-y)^2/2t$ nor u.) Equivalently $Q = \max_y \int_{-\infty}^{y} (-u_0)$ and $P = \max_y \int_y^{\infty} u_0$, so in general $P \leq \int u_0^+$ and $Q \leq \int u_0^-$, with equality exactly when $u_0 \leq 0$ to the left and $u_0 \geq 0$ to the right of some point (a "pre-formed" N-shape, with no cancellation). Example 8.15 has $V \geq 0$, so $Q = 0$ and $P = M - 0 = 0$: the limiting N-wave is $\mathcal{N} \equiv 0$, consistent with $\|u\|_{L^1} = 1/t \to 0$.

Heuristic derivation of the constants (not a proof of convergence). The shock-merging lemma and the computation below *motivate* the form of the theorem and the formula for P and Q; they do not establish the L^1 convergence, which requires the estimates in the cited sources. The constants (58) arise from the Lax–Oleinik formula (43) of Remark 8.5 by a short computation. Here $V(y) = 0$ to the left of the support, $V(y) = M$ to its right, and $\min V = -Q$ is attained at some bounded point y_0. For x of order $\sqrt{t}$ the minimiser of $V(y) + (x-y)^2/(2t)$ is one of two candidates:

- $y \approx x$, with value 0 for x left of the support and M for x right of it, giving $u = (x-y)/t = 0$;
- $y \approx y_0$, with value $-Q + (x-y_0)^2/(2t) \approx -Q + x^2/(2t)$, giving $u = (x-y_0)/t \approx x/t$.

The second beats the first on the left when $-Q + x^2/(2t) < 0$, i.e. $x > -\sqrt{2Qt}$, and on the right when $-Q + x^2/(2t) < M$, i.e. $x < \sqrt{2(M+Q)t} = \sqrt{2Pt}$. This produces exactly the profile (57), with shocks (ties between the two candidates) at $x = -\sqrt{2Qt}$ and $x = \sqrt{2Pt}$. Making this rigorous — controlling the error in L^1 — is the content of the cited theorem, and is not attempted here. In this picture the conserved quantity is only $\int u\,dx = M = P - Q$; the individual masses P and Q are *not* conserved, since positive and negative parts cancel at shocks as in Example 8.15.

8.6.4 A numerical check of the constants

The following is an *observation*, not a proof. For piecewise-constant compactly supported u_0 the entropy solution can be computed (to grid resolution) directly from the Lax–Oleinik formula (43), with V the exact primitive (`code/nwave_check.py`). We compare it in L^1 with the N-wave built from the corrected constants (58) and, for contrast, from the *naive* constants $P = \int u_0^+$, $Q = \int u_0^-$. Three data sets were used: (a) -1 on $(0,1)$, $+1$ on $(1,2)$ (no cancellation); (b) $+1$ on $(0,1)$, -1 on $(1,2)$ (Example 8.15); (c) six alternating-sign pieces with values $1.2, -0.8, 1.6, -1.4, 0.9, -0.6$ on $(0, 0.7, 1.5, 2.1, 3.0, 3.8, 4.4)$.

data	corrected (P,Q)	naive (P,Q)	L^1 error at $t = 6400$	naive error	fitted slope
(a)	$(1,1)$	$(1,1)$	7.0×10^{-2}	7.0×10^{-2}	-0.49
(b)	$(0,0)$	$(1,1)$	1.6×10^{-4}	2.0	-1.00
(c)	$(0.36, 0.10)$	$(2.52, 2.26)$	9.4×10^{-2}	4.4	-0.44

Table 5: L^1 distance between the entropy solution and the N-wave, for $t \in \{25, 100, 400, 1600, 6400\}$; "fitted slope" is the log–log slope of the corrected-constant error over $t \geq 100$.

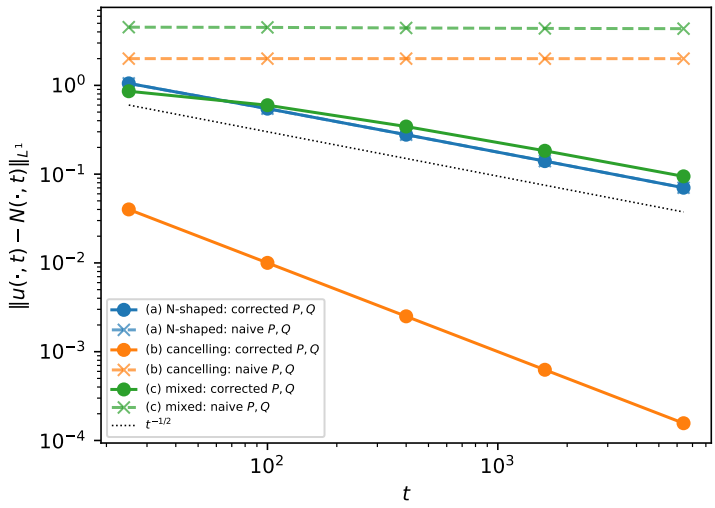


Figure 6: L^1 distance to the N-wave against t: corrected constants (solid) decay; naive constants (dashed) do not in cases (b) and (c).

With the corrected constants the error decays in all three cases. In (a) the slope is -0.49, consistent with $t^{-1/2}$; in (b) the limiting N-wave is $\mathcal{N} \equiv 0$ and the error equals $\|u(\cdot,t)\|_{L^1} = 1/t$ exactly (slope -1.00, as computed in Example 8.15); in (c) the slope over this range is -0.44 and still changing, so no rate is claimed there. With the naive constants the error does not decay in (b) or (c): it stays near $\|\mathcal{N}_{\text{naive}} - \mathcal{N}_{\text{corr}}\|_{L^1} = (P_{\text{n}} - P_{\text{c}}) + (Q_{\text{n}} - Q_{\text{c}})$ (the extra mass of the larger naive wave, which has the same profile on the common interval), namely 2.0 in (b) and $2.16 + 2.16 = 4.32$ in (c) (observed 4.35 at $t = 6400$). The naive profile itself has L^1 norm $P + Q = 2.0$ and 4.78 respectively; in (b) the two quantities coincide, which can hide the distinction. Either way, the positive and negative masses of u_0 are not the correct constants.

9 Conclusion

The answer to the guiding question. The question of Section 1.1 was why the inviscid Burgers' equation needs an entropy condition and why that condition is the right one. The chain of results is now complete. Smooth decreasing data develop an infinite slope at an explicit time and place, $T^* = -1/\min u_0'$ (Section 3); the weak formulation continues the solution past this point through jumps obeying the Rankine–Hugoniot condition (Section 4), but it is non-unique, already for the simplest Riemann data, where in fact infinitely many weak solutions exist (Section 5). The Oleinik entropy condition, of which the Lax condition $u_\ell > u_r$ is the local form at a jump, restores uniqueness and gives the complete solution of the Riemann problem (Sections 6–7). That it is the *right* condition is the content of the vanishing viscosity limit: via the Cole–Hopf transformation the viscous solutions converge to the entropy solution,

whose jumps can only be downward because the Lax–Oleinik minimiser is monotone, and directly because a viscous travelling wave connecting u_ℓ to u_r exists exactly when $u_\ell > u_r$ (Section 8, Proposition 8.6). The theory itself is standard [5, §3.4]; the novelty of this paper is in the explicit verifications and the computations below.

What the computations added.

- *An explicit example.* For $u_0 = -\arctan x$ the entropy solution after breakdown is given in closed parametric form, with a single stationary shock at $x = 0$ of strength $\arctan y^*(t)$, where y^* is the positive root of $\arctan y = y/t$. The strength grows like $\sqrt{3(t-1)}$ from the breakdown time $t = 1$ and saturates at $\pi/2$ (Section 8.4).
- *A prediction that was confirmed.* A viscous shock has L^1 distance exactly $4\ln 2\,\nu \approx 2.77\,\nu$ from the sharp shock, independent of its strength and speed. In the Cole–Hopf solutions the shock layer contributes $2.78\,\nu$, within 0.5% of this, and the contribution tends to $4\ln 2\,\nu$ at every strength tested (0.52–1.50); it is visibly larger in a wide window only for young shocks, where the steep outer solution contaminates it. The total L^1 error is first order in ν, tending to $\approx 3.11\,\nu$, the remaining $\approx 0.33\,\nu$ coming from the smooth part (which is measured, not derived).
- *Godunov as a vanishing viscosity.* The scheme converges at first order for the shock, with error exactly $0.50\,\Delta x$ when the shock lies on a cell interface and 0.6–$0.77\,\Delta x$ for the misaligned grids tested; for the rarefaction the observed order is about 0.8 and still rising, so a first-order rate is not established there. The modified equation predicts the shock error to within 3% at CFL 0.45 (and within about 10% for CFL numbers from 0.1 to 0.95), corresponding to an effective viscosity $\approx 0.18\,\Delta x \to 0$: a monotone scheme selects the entropy solution because its numerical diffusion acts as a vanishing viscosity. This is a heuristic consistency check; the rigorous result is the convergence theorem for monotone schemes [4].
- *Corrected N-wave constants.* The long-time N-wave is determined by $Q = -\min V$ and $P = \int u_0 - \min V$, where V is the primitive of u_0, *not* by the positive and negative masses of u_0: positive and negative parts cancel across shocks, and only $P - Q = \int u_0$ is conserved. For the data $+1$ on $(0,1)$, -1 on $(1,2)$ the correct limit is 0 (the L^1 norm decays exactly like $1/t$), while the positive/negative masses would predict a wave of norm 2. Numerically the distance to the N-wave with the correct constants decays in every case tested, and with the naive constants it does not in the cases where the two differ (Section 8.6).

Limits of the exposition. The proofs of the uniqueness theorem (Theorem 6.7), of the vanishing-viscosity limit (Theorem 8.4), of the convergence of the Godunov scheme (Theorem 8.10) and of the N-wave decay (Theorem 8.16) are cited, not reproduced. The numerical statements are observations for the specific data sets and resolutions stated. Burgers' equation is a model of nonlinear steepening and its regularisation, not a stepping stone to the far harder open regularity questions for the three-dimensional Euler and Navier–Stokes equations, whose mechanisms differ.

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